

Build Your First AI Voice Agent

with the GPT Realtime API

A voice assistant that listens, thinks, calls APIs and answers in natural speech.

AUG 4 · 11:00 AM

2-2.5 hours · LIVE



Real-time voice



GPT Realtime API



AI tool calling



Integrate APIs



Conversational
memory

POWERED BY



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Engineering Manager at Swiggy — leading the Agentic Engineering team

Agentic systems in production

Voice & realtime AI

Mentoring engineers



11 years
of engineering experience

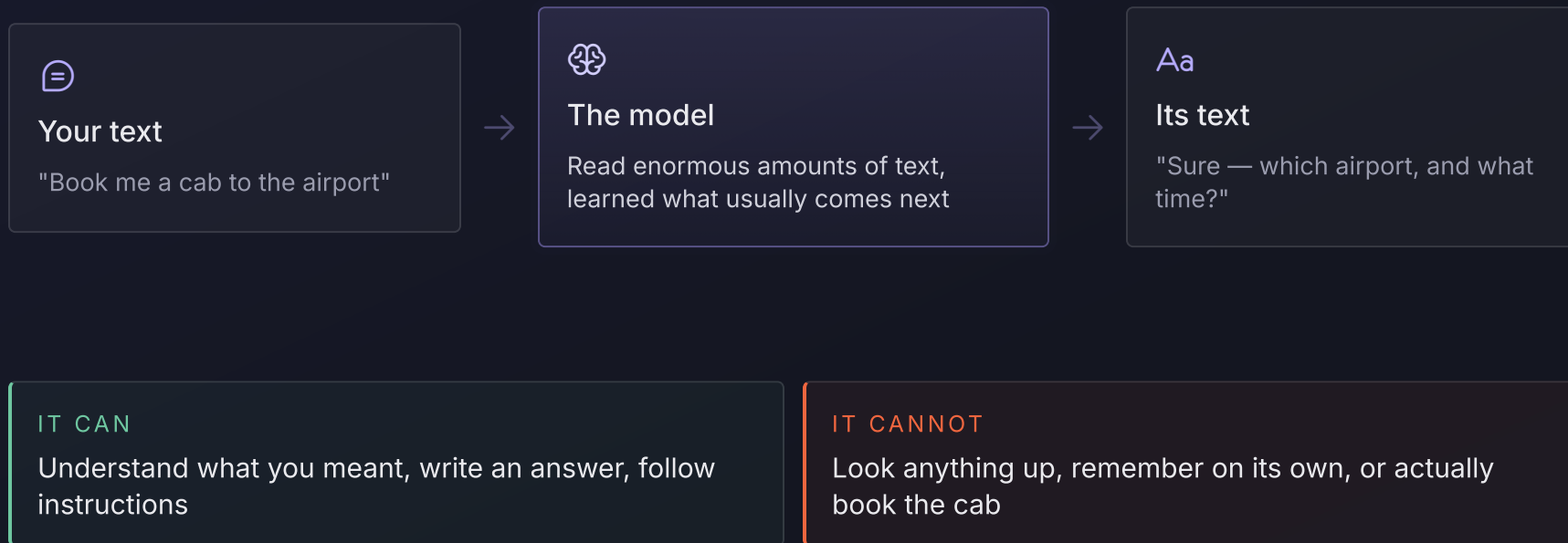


Founder & Mentor
Rebind Rise

- FOUNDATIONS

What is an LLM?

Text goes in. Text comes out. That's the whole contract.



- FOUNDATIONS

What is a token?

A chunk of text — usually a word, sometimes a piece of one.

ROUGH RULE

1 token $\approx$ 4 characters

WHAT YOU TYPE

Book me a cab to the airport

↓ split into tokens

Book	me	a	cab	to	the	air	port
3841	665	261	7402	316	290	1247	3106

The model never sees letters or sentences — only these numbers. Tokens are also the unit you're billed in, and the unit latency is measured in.

- FOUNDATIONS

An LLM only predicts the next token

One token at a time — then it reads its own output and does it again.

GENERATING

Sure — which airport are you flying from |

↩ every token it writes is fed back in as input for the next one

UNDER THE HOOD, EACH STEP

"...which ____"

airport		74%
city		18%
terminal		8%

It scores every possible token and picks one. That's the entire move.



Replies stream — you see word one before the last is decided



Longer answer = more steps = more time



A confident guess can still be wrong

• FOUNDATIONS

A brain with no hands

It can decide what to do. It can't do it. So we give it tools.



Can't see live data — prices, seats, your database



Can't change anything — no booking, no payment, no email



Asked anyway, it invents a confident answer



Model decides

"call get_cab_fare(to: BOM)"



Your code runs it

a normal Python function



Real system answers

API, DB, payment gateway



Result back to model

"₹840" → it replies in words



The model never executes anything itself. It only names a tool and its arguments — you stay in control of what actually runs.

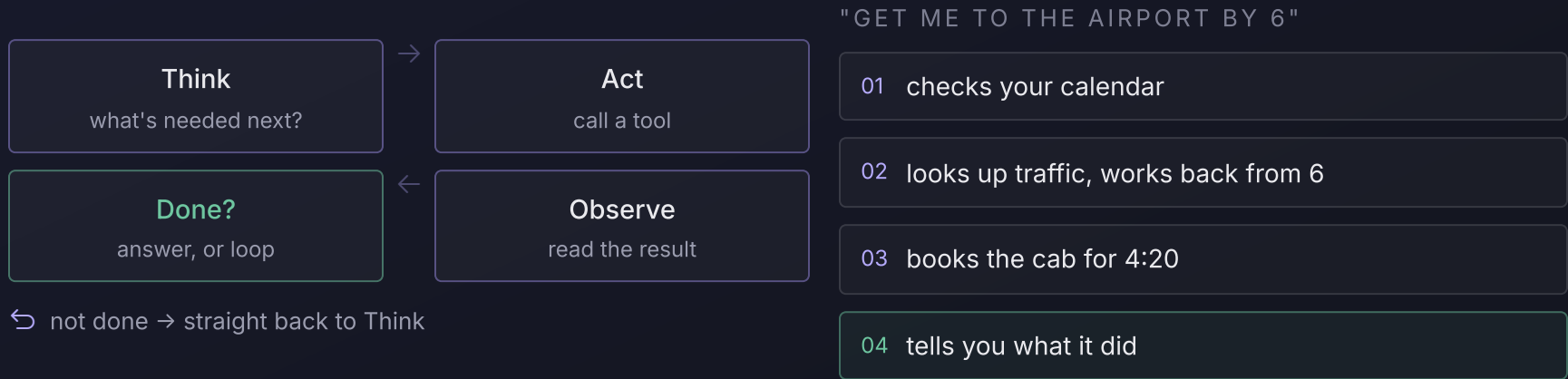
- FOUNDATIONS

So what is an agent?

A model in a loop, with tools and a goal — it keeps going until it's done.

chatbot = one reply

agent = keeps working



Today we build this loop with speech on both ends — and the next three slides are the three ways to wire it up.

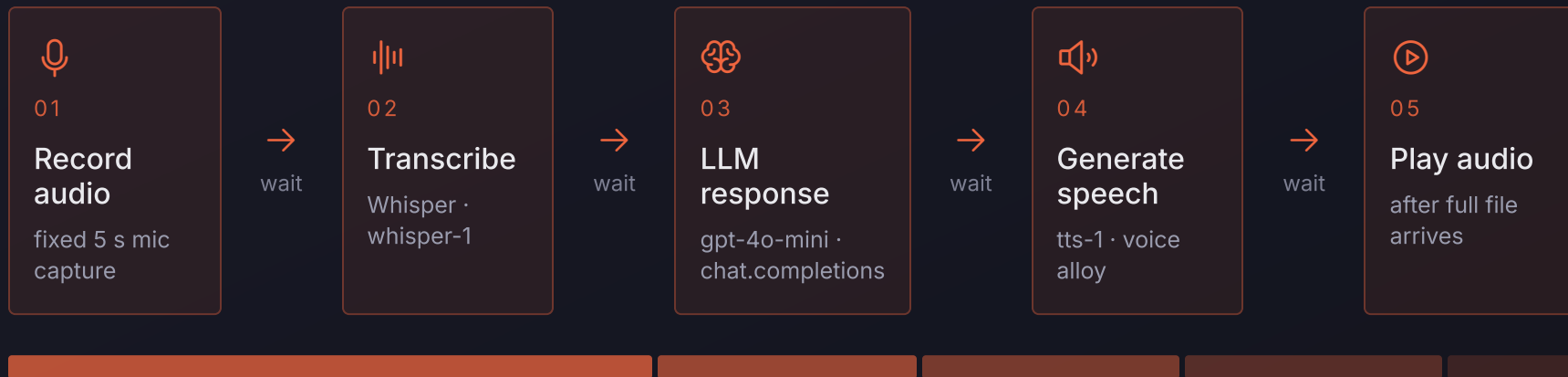
• ARCHITECTURE 01

Naive Sequential Pipeline

STT → LLM → TTS, one blocking call at a time

TIME TO FIRST SOUND

4–6 s



Serial timeline — no stage overlaps the one before it



Failure mode: silence still transcribes — Whisper hallucinates a phrase, and the LLM + TTS calls are spent answering it.

Optimized Streaming Pipeline

Same three models — stages overlap instead of queueing

PERCEIVED RESPONSE

400–900 ms

Mic stream

continuous over WebSocket



Streaming STT + server VAD

gpt-4o-transcribe · partials every 100–300 ms



Turn-end detection

speech_started / completed events



Incremental LLM generation

gpt-4o-mini · token-by-token



Streaming TTS

tts-1 · alloy · chunks as generated



Playback

starts before synthesis finishes



user speaking

turn end

first audio out



Playback loops straight back into listening — one continuous multi-turn session, no restart per turn.

Realtime Speech-to-Speech

API HOPS PER TURN

0

Browser-direct WebRTC — the backend leaves the audio path

BROWSER

Opens a direct WebRTC peer connection with the ephemeral secret

mic track out · model audio in · data channel for events

audio ⇌ audio · WebRTC



bypasses backend

OPENAI REALTIME

One persistent session: audio in → reasoning → audio out

gpt-realtime-2.1 · voice marin

transcripts via gpt-4o-mini-transcribe

interruption / barge-in built in

BACKEND · FASTAPI

POST /realtime/session mints a client secret

no audio ever passes through



POST /v1/realtime/
client_secrets

TOOL-CALLING LOOP

function_call
event



runToolCall in browser
JS



POST
/tools/call



function_call_output



response.create → keeps
speaking

Comparison

slow  fast

Naive sequential

LATENCY

4–6 s

API HOPS PER TURN

3 blocking calls

INTERRUPTION

None

COMPLEXITY

Lowest — 40 lines

Optimized streaming

LATENCY

400–900 ms

API HOPS PER TURN

3, overlapped

INTERRUPTION

Partial — VAD driven

COMPLEXITY

Highest — you own the orchestration

Realtime speech-to-speech

LATENCY

Near-instant

API HOPS PER TURN

0 — one live session

INTERRUPTION

Native barge-in

COMPLEXITY

Moderate — secrets + tool bridge

• IN THE WILD

Where voice agents are shipping

NEEDS

Whenever hands are busy, the queue is long, or typing is slower than talking.

realtime + tool calling



Customer support

Order status, refunds, rescheduling — resolved without a queue



Outbound calling

Appointment reminders, lead qualification, payment follow-ups



Clinical scribing

Doctor talks, notes get written — no keyboard in the room



Drive-thru & kiosks

Takes the order, upsells, handles "actually, make that two"



Field & warehouse

Hands on the job, voice does the data entry and lookups



Tutors & interviews

Language practice, mock interviews, spoken assessments



People interrupt — barge-in isn't optional



Value lives in the tools it can call



Past ~1s of silence, it stops feeling human

• COHORT HAPPENING NOW



Agentic AI Engineering Bootcamp

10 weeks, in person in Hinjawadi, Pune — for working software engineers & students*

RAG & vector DBs

Tool calling

loops & AI agents

Streaming & realtime AI

MCP servers and client

Multi Agent deployment and harness

From basic LLM tokens to voice building voice agent

FORMAT

Weekends only

COHORT SIZE

Max 25 seats

CAPSTONE

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